

# PAPER\_279 — Sombrero SMBH Dominance Ratio $\gamma_{\text{BH}}$ and UQFF Sphere of Influence $r_{\text{SOI}}$

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## Abstract

The Sombrero Galaxy (M104) harbours one of the most massive black holes relative to its host galaxy mass of any object in the Local Universe:  $M_{\text{BH}} = 10^9 M_{\text{sun}}$  in a galaxy of total mass  $M = 10^{11} M_{\text{sun}}$ , giving a **SMBH Dominance Ratio**  $\gamma_{\text{BH}} = M_{\text{BH}}/M = 0.01$  (1%). For comparison, the Milky Way's Sgr A\* has  $\gamma_{\text{BH}} \approx 4 \times 10^{-5}$  (~0.004%); Sombrero's SMBH is **250× more dominant relative to its host**. Within the UQFF framework, we define the **UQFF Sphere of Influence**  $r_{\text{SOI}} = r \times \sqrt{(\gamma_{\text{BH}})}$ , the radius at which the central BH's direct gravitational contribution equals the total galaxy contribution at the reference radius. For Sombrero,  $r_{\text{SOI}} = 2.36 \times 10^{19} \text{ m}$  — a precise UQFF prediction setting the boundary inside which BH gravity exceeds galaxy-mean gravity in the UQFF model.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis  $\blacklozenge$  establishing a new connection in the UQFF framework not present in Standard Model treatments.

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## 1. Observational Motivation

### 1.1 The Sombrero SMBH

The Sombrero Galaxy's central black hole mass has been measured through stellar and gas kinematics:

- Ford et al. (1996):  $M_{\text{BH}} = (1.0 \pm 0.75) \times 10^9 M_{\text{sun}}$  (gas kinematics, HST)
- Kormendy et al. (1996): confirmation from stellar dynamics
- Jardel et al. (2011):  $M_{\text{BH}} \approx 6.6 \times 10^8 M_{\text{sun}}$  (JAM modelling, consistent with  $\sim 10^9$  range)

We adopt  $M_{\text{BH}} = 1.0 \times 10^9 M_{\text{sun}} = 1.989 \times 10^{39} \text{ kg}$ .

The total galaxy mass (stars + gas + DM within the reference radius) is  $M = 10^{11} M_{\text{sun}} = 1.989 \times 10^{41} \text{ kg}$ .

### 1.2 Why $\gamma_{\text{BH}}$ Matters

In standard astrophysical models, BH sphere-of-influence calculations use the velocity dispersion ( $r_{\text{SOI}} = GM_{\text{BH}}/\sigma^2$ ). The UQFF framework generalises this to a mass-ratio-based definition consistent with the 26-layer Triadic gravity structure, yielding a dimensionless parameter  $\gamma_{\text{BH}}$  that naturally encodes the BH's dominance within the UQFF vacuum field hierarchy.

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## 2. Theoretical Derivation

### 2.1 SMBH Dominance Ratio

We define the dimensionless **SMBH Dominance Ratio**:

$$\gamma_{\text{BH}} = \frac{M_{\text{BH}}}{M}$$

For the Sombrero Galaxy:

$$\gamma_{\text{BH}}^{\text{Sombrero}} = \frac{10^9 M_{\odot}}{10^{11} M_{\odot}} = 0.01 = 1\%$$

### 2.2 SMBH Direct Gravitational Contribution

The BH contribution to  $g_{\text{total}}$  at the reference radius  $r$ :

$$g_{\text{BH}} = \frac{GM_{\text{BH}}}{r^2} = \gamma_{\text{BH}} \cdot \frac{GM}{r^2} = \gamma_{\text{BH}} \cdot g_{\text{base}}$$

Substituting  $\gamma_{\text{BH}} = 0.01$  and  $g_{\text{base}} = 2.382 \times 10^{-10} \text{ m/s}^2$ :

$$g_{\text{BH}} = 0.01 \times 2.382 \times 10^{-10} = 2.382 \times 10^{-12} \text{ m/s}^2$$

### 2.3 UQFF Sphere of Influence

The **UQFF Sphere of Influence**  $r_{\text{SOI}}$  is defined as the radius  $r' < r$  where the BH gravitational contribution equals the total galaxy contribution at  $r$ :

$$g_{\text{BH}}(r') = g_{\text{base}}(r)$$

$$\frac{GM_{\text{BH}}}{r'^2} = \frac{GM}{r^2}$$

Solving:

$$r'^2 = r^2 \cdot \frac{M_{\text{BH}}}{M} = r^2 \cdot \gamma_{\text{BH}}$$

$$r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$$

For Sombrero:

$$r_{\text{SOI}} = 2.36 \times 10^{20} \cdot \sqrt{0.01} = 2.36 \times 10^{20} \times 0.1 = 2.36 \times 10^{19} \text{ m}$$

**Physical interpretation:** Within  $r_{\text{SOI}} = 2.36 \times 10^{19} \text{ m}$  ( $\sim 2.49 \text{ kly}$ ), the direct BH gravitational acceleration exceeds the galaxy-mean  $g_{\text{base}}$ . This is the UQFF-predicted boundary of BH gravitational dominance.

## 2.4 Verification

At  $r' = r_{\text{SOI}}$ :

$$g_{\text{BH}}(r_{\text{SOI}}) = \frac{GM_{\text{BH}}}{r_{\text{SOI}}^2} = \frac{G \cdot 0.01M}{(0.1r)^2} = \frac{0.01 \cdot GM}{0.01 \cdot r^2} = \frac{GM}{r^2} = g_{\text{base}} \checkmark$$


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## 3. Comparative SMBH Dominance Table

| Galaxy                 | M_BH (M_sun)                  | M_total (M_sun)                  | $\gamma_{\text{BH}}$      | r_SOI / r_ref             |
|------------------------|-------------------------------|----------------------------------|---------------------------|---------------------------|
| Milky Way (Sgr A*)     | $\sim 4 \times 10^6$          | $\sim 10^{11}$                   | $\sim 4 \times 10^{-5}$   | $\sim 6.3 \times 10^{-3}$ |
| Andromeda (M31)        | $\sim 1.4 \times 10^8$        | $\sim 10^{12}$                   | $\sim 1.4 \times 10^{-4}$ | $\sim 1.2 \times 10^{-2}$ |
| M87                    | $\sim 6.5 \times 10^9$        | $\sim 6 \times 10^{12}$          | $\sim 1.1 \times 10^{-3}$ | $\sim 3.3 \times 10^{-2}$ |
| <b>Sombrero (M104)</b> | <b><math>\sim 10^9</math></b> | <b><math>\sim 10^{11}</math></b> | <b>0.01</b>               | <b>0.1</b>                |

**Sombrero's  $\gamma_{\text{BH}} = 0.01$  is the highest of any nearby well-measured galaxy in the UQFF catalogue, making it the dominant test-case for SMBH-galaxy UQFF coupling.**

Key comparison ratios: - Sombrero/Sgr A\*:  $\gamma_{\text{BH}}$  ratio =  $0.01/4 \times 10^{-5} = \mathbf{250\times}$  - Sombrero/M87:  $\gamma_{\text{BH}}$  ratio =  $0.01/1.1 \times 10^{-3} \approx \mathbf{9\times}$  - Sombrero/Andromeda:  $\gamma_{\text{BH}}$  ratio =  $0.01/1.4 \times 10^{-4} \approx \mathbf{71\times}$

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## 4. UQFF BH Contribution in the Master Equation

The BH term enters computeG() as an additive contribution alongside the 26-layer Triadic sum:

$$g_{\text{total}} = [\dots + g_{\text{BH}} + \dots] \cdot \kappa_{\text{recession}} \cdot \sigma_{\text{SC}}$$

$$= [\dots + 2.382 \times 10^{-12} \text{ m/s}^2 + \dots] \times 0.99374 \times (1 - 10^{-20})$$

**BH fractional contribution to  $g_{\text{total}}$  (estimated):**

$$\frac{g_{\text{BH}}}{g_{\text{total}}} \approx \frac{2.382 \times 10^{-12}}{1.238 \times 10^{-8} + 2.382 \times 10^{-12} + \dots} \approx \frac{2.382 \times 10^{-12}}{1.24 \times 10^{-8}} \approx 1.9 \times 10^{-4}$$

While the BH's direct gravitational contribution at the reference radius  $r$  is a small fraction of the 26-layer Triadic sum ( $\sim 0.019\%$ ), the  $r_{\text{SOI}}$  formula reveals that inside  $2.36 \times 10^{19}$  m, BH gravity dominates the reference-point baseline — a qualitative distinction for UQFF predictions of inner galactic structure.

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## 5. Module Implementation

```
// PAPER_279 – SOMBRERO_UQFF_MODULE.cpp, updateCache()
gamma_BH = M_BH / M; // 0.01 = 1%
r_SOI = r * std::sqrt(gamma_BH); // 2.36e20 × 0.1 = 2.36e19 m

// Applied in computeG():
double g_BH = G_grav * M_BH / (r * r); // = gamma_BH * g_base = 2.382e-12 m/s2
```

#### Computed values for Sombrero M104:

| Quantity                                             | Value                   | Units            |
|------------------------------------------------------|-------------------------|------------------|
| $M_{\text{BH}} = 10^9 M_{\text{sun}}$                | $1.989 \times 10^{39}$  | kg               |
| $M = 10^{11} M_{\text{sun}}$                         | $1.989 \times 10^{41}$  | kg               |
| $\gamma_{\text{BH}} = M_{\text{BH}}/M$               | 0.01                    | dimensionless    |
| $g_{\text{BH}} = G \cdot M_{\text{BH}}/r^2$          | $2.382 \times 10^{-12}$ | m/s <sup>2</sup> |
| $r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$ | $2.36 \times 10^{19}$   | m                |
| $r_{\text{SOI}}$ in kly                              | ~2.49                   | kly              |

## 6. Key Constants Introduced

| Symbol               | Value                   | Units            | Description                                  |
|----------------------|-------------------------|------------------|----------------------------------------------|
| $\gamma_{\text{BH}}$ | 0.01                    | dimensionless    | SMBH Dominance Ratio<br>$M_{\text{BH}}/M$    |
| $r_{\text{SOI}}$     | $2.36 \times 10^{19}$   | m                | UQFF Sphere of Influence<br>radius           |
| $g_{\text{BH}}$      | $2.382 \times 10^{-12}$ | m/s <sup>2</sup> | BH direct gravitational<br>contribution at r |

## 7. Physical Significance

- SMBH Dominance Ratio as a universal UQFF parameter:**  $\gamma_{\text{BH}} = M_{\text{BH}}/M$  provides a single dimensionless number characterising how BH-dominated a galaxy is. It ranges from  $\sim 10^{-5}$  (Sgr A\*) to  $\sim 0.01$  (Sombrero), spanning three orders of magnitude in the current UQFF catalogue.
- UQFF Sphere of Influence formula:**  $r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$  is derived directly from setting  $g_{\text{BH}}(r') = g_{\text{base}}(r)$ . This provides a clean, parameter-free prediction for the BH dominance scale in any UQFF module with a known  $M_{\text{BH}}/M$  ratio.
- Sombrero as extreme BH-galaxy system:** With  $\gamma_{\text{BH}}$  250× larger than the Milky Way's, Sombrero tests UQFF behaviour in the BH-dominated regime. The large  $\gamma_{\text{BH}}$  implies that inner BH effects begin influencing the 26-layer Triadic structure at radii as large as  $r_{\text{SOI}} = 2.36 \times 10^{19}$  m.
- AGN feedback implications:** The large  $M_{\text{BH}}$  implies strong AGN feedback potential. UQFF predicts that AGN activity modifies the vacuum energy density locally, which would appear in the UQFF framework as a time-varying component of  $Ug1\_vec[i]$  for the innermost layers — a direction for future UQFF module development.

5. **Generalisation:** The formula  $\gamma_{\text{BH}} = M_{\text{BH}}/M$  and  $r_{\text{SOI}} = r_{\sqrt{(\gamma_{\text{BH}})}}$  define a universal UQFF BH dominance prescription applicable to any galaxy module. Future UQFF modules for BH-dominated systems (e.g., NGC 1277, M87) should include this pair of parameters as standard.
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## 8. References

- Ford, H. C. et al. (1996). ApJ, 458, 132. (Sombrero  $M_{\text{BH}}$  from gas kinematics, HST)
  - Kormendy, J. et al. (1996). ApJ, 459, L57. (Sombrero BH mass confirmation)
  - Jarrel, J. R. et al. (2011). ApJ, 739, 21. (Sombrero DM halo and BH mass)
  - Gültekin, K. et al. (2009). ApJ, 698, 198. ( $M_{\text{BH}}-\sigma$  correlation)
  - PAPER\_277: UQFF Gravitational Recession Damping  $\kappa_{\text{recession}}$  for  $z = +0.0063$
  - PAPER\_278: Sombrero Dust Ring UQFF Gravitational Ring Resonator  $\omega_{\text{ring}}$
  - SOMBRERO\_UQFF\_MODULE.cpp (UQFF 2.0, Session 77)
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*UQFF 2.0 — The SMBH Dominance Ratio  $\gamma_{\text{BH}} = M_{\text{BH}}/M$  and UQFF Sphere of Influence  $r_{\text{SOI}} = r_{\sqrt{(\gamma_{\text{BH}})}}$  are new universal parameters for UQFF galaxy modules, first derived and tested on the Sombrero Galaxy. — Daniel T. Murphy, Session 77, March 2026.*

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## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_{\mu}\phi_B)(\partial^{\mu}\phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.135 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 29, \quad n_{\text{channel}} = 20/26$$

Since  $p_{\text{DVP}} = 29$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10<sup>3</sup> yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

## §B.4 Production-Scale Consistency

| Framework            | Canonical Value                                  | This Paper                                                     | Status                              |
|----------------------|--------------------------------------------------|----------------------------------------------------------------|-------------------------------------|
| VDS ratio            | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$     | Local sub-ratio = 0.135                                        | ✓ Threshold-consistent              |
| DVP prime BSH layers | $p_k \in \{2,3,\dots,113\}$<br>26 harmonic terms | $p_{\text{DVP}} = 29$<br>$j = 1\dots 26,$<br>$\cos(2\pi j/26)$ | ✓ Resonant<br>✓ Full 26D projection |
| $\kappa$ decay       | $5.0 \times 10^{-4} \text{ day}^{-1}$            | Applied in VDS exponential                                     | ✓ Canonical                         |
| [SSq]                | 0.57                                             | Applied in BSH saturation                                      | ✓ Canonical                         |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                        | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|------------------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                       | 1/137.036                              | PDG 2024                            | ✓ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)                | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | ✓ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | ✓ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                               | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*